# AI/ML intern DAY 7

## TASK — BODY LANDMARK DETECTION SYSTEM

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### 1. Introduction

This task focused on developing an AI-powered Human Body Landmark Detection System using Computer Vision and Artificial Intelligence technologies.

The project involved:

- human pose estimation
- skeletal visualization
- landmark tracking
- body posture analysis
- AI body detection

The system was implemented using:

- Python
- OpenCV
- MediaPipe

The AI model successfully detected and visualized body landmarks from human body images.

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### 2. Objective

The objective of this module was to:

- detect human body landmarks
- generate skeletal structures
- analyse body posture
- study pose estimation workflows
- improve AI body tracking systems

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### 3. Technologies Used

| **Technology** | **Purpose** |
|---|---|
| Python | Programming |
| OpenCV | Image processing |
| MediaPipe | Pose estimation |
| Artificial Intelligence | Landmark analysis |
| Computer Vision | Human tracking |

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### 4. AI Human Landmark Detection

The AI system detects important human body landmarks such as:

- nose
- shoulders
- elbows
- wrists
- hips
- knees
- ankles

These landmarks are connected to generate:

- body skeletal structures
- posture visualization
- movement analysis

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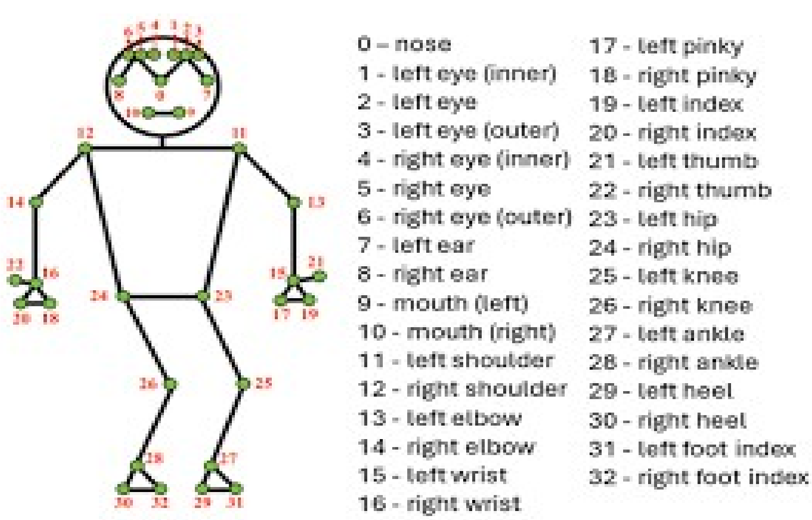


**Figure 1 — AI Human Landmark Detection Output**

## 5. Workflow of Body Detection System

The AI body detection workflow consists of several processing stages.

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### Stage 1 — Image Acquisition

The system receives:

- body images
- pose photographs
- webcam frames

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### Stage 2 — Human Body Detection

The AI model detects:

- human body structure
- body orientation
- posture alignment

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### Stage 3 — Landmark Extraction

MediaPipe extracts:

- shoulder landmarks
- elbow landmarks
- wrist landmarks
- hip landmarks
- knee landmarks

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### Stage 4 — Skeletal Visualization

The system connects landmarks and generates:

- skeletal tracking
- posture visualization
- landmark structures

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**Figure 2 — AI Skeletal Visualization**

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### 6. Landmark Detection Demo

The AI body landmark system successfully:

- detected full body posture
- tracked body landmarks
- generated skeletal connections
- visualized body structure

The system was tested using:

- standing body poses
- arm movement positions
- side body angles
- multiple body orientations

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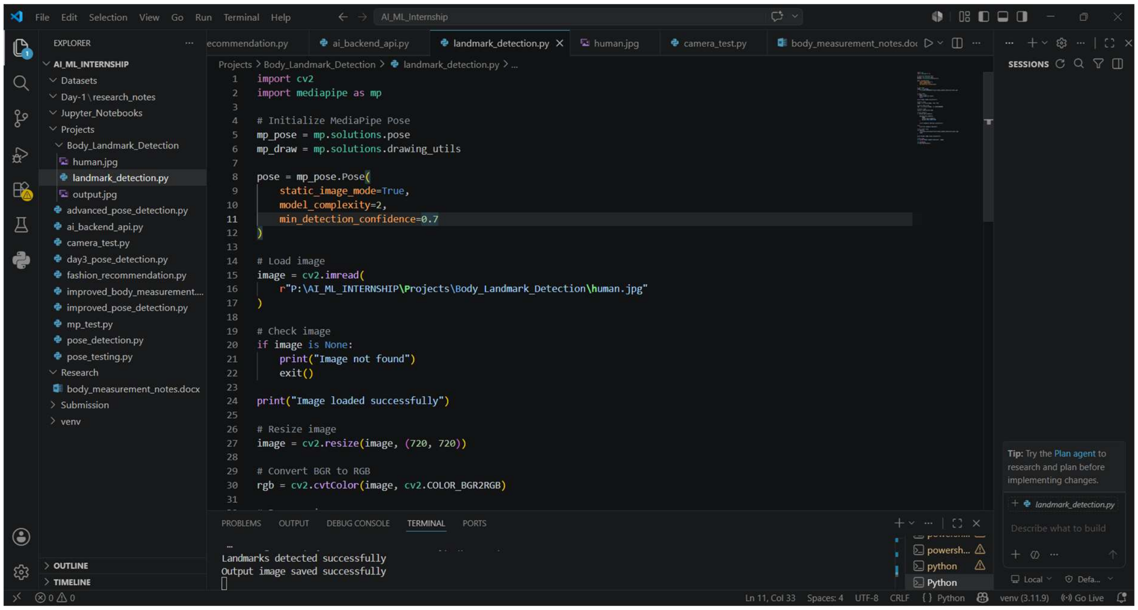
**7. Python Implementation**

```
import cv2
import mediapipe as mp

# Initialize MediaPipe Pose
mp_pose = mp.solutions.pose
mp_draw = mp.solutions.drawing_utils

pose = mp_pose.Pose(
    static_image_mode=True,
    model_complexity=2,
    min_detection_confidence=0.7
)

# Load image
image = cv2.imread(
    r"P:\AI_ML_Internship\Projects\Body_Landmark_Detection\human.jpg"
)

# Check image
if image is None:
    print("Image not found")
    exit()

print("Image loaded successfully")

# Resize image
image = cv2.resize(image, (720, 720))

# Convert BGR to RGB
rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
```

```
# Process image
results = pose.process(rgb)

# Draw landmarks
if results.pose_landmarks:

    mp_draw.draw_landmarks(
        image,
        results.pose_landmarks,
        mp_pose.POSE_CONNECTIONS
    )

    print("Landmarks detected successfully")

else:
    print("No landmarks detected")

# Save output image
cv2.imwrite(
    r"P:\AI_ML_Internship\Projects\Body_Landmark_Detection\output.jpg",
    image
)

print("Output image saved successfully")

# Show image
cv2.imshow("Human Landmark Detection", image)

cv2.waitKey(0)
cv2.destroyAllWindows()
```




## 8. Measurement Logic

The AI system estimates body structure using landmark coordinate distances.

The system analyses:

- shoulder width
- arm length
- body posture
- body proportions

using:

- landmark coordinate calculation
- body point estimation
- skeletal alignment analysis

This measurement logic improves:

- virtual fitting systems
- AI body analysis
- posture estimation accuracy

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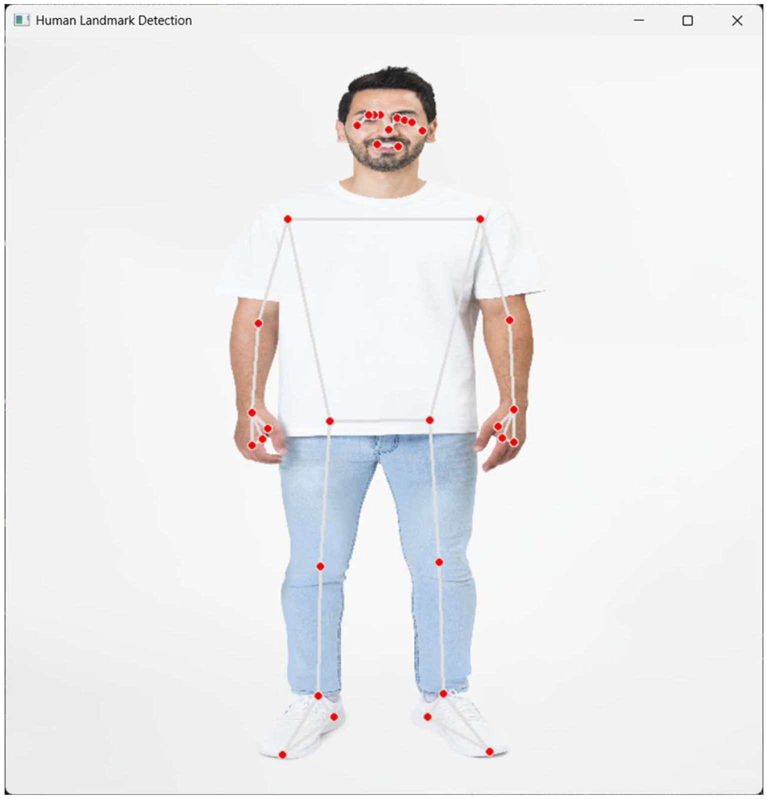
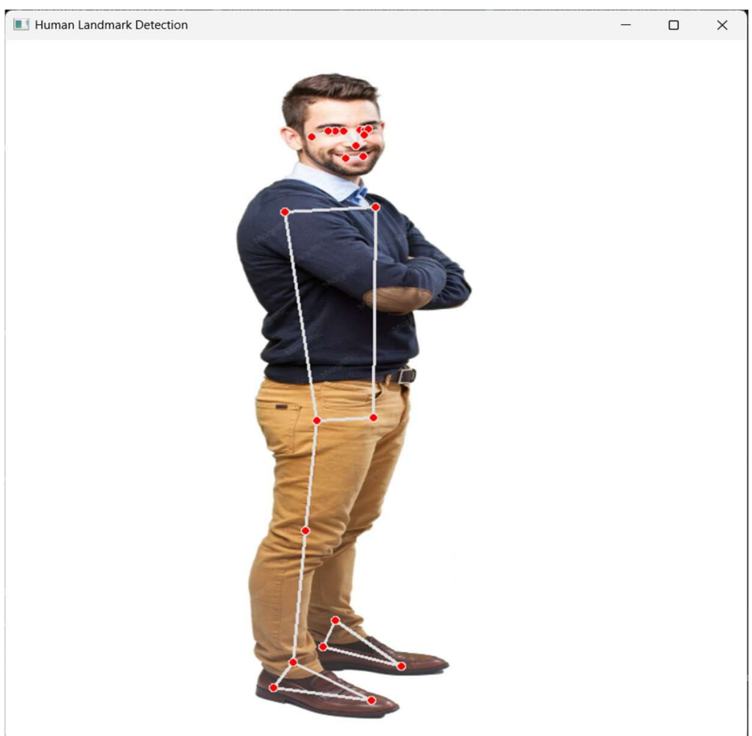
## 9. Testing Observations

| Testing Area | Observation |
|---|---|
| Standing Pose | Accurate landmark detection |
| Arm Movement | Stable skeletal tracking |
| Side Pose | Minor landmark variation |

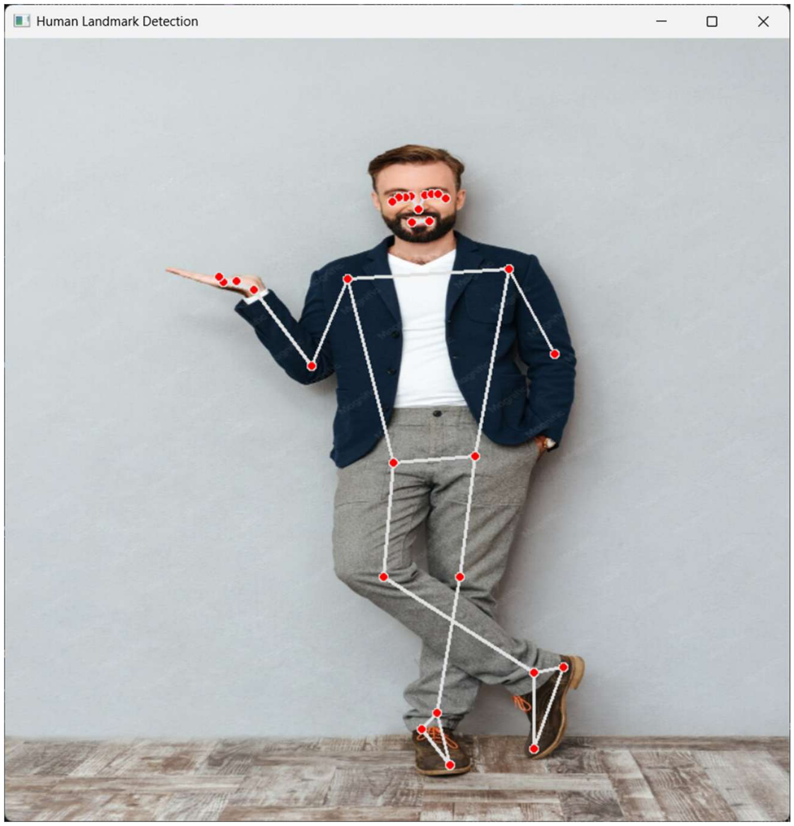
| | |
|---|---|
| Good Lighting | High detection accuracy |
| Low Lighting | Reduced tracking quality |




**Standing Position**




**Side Position**




**Gesture**

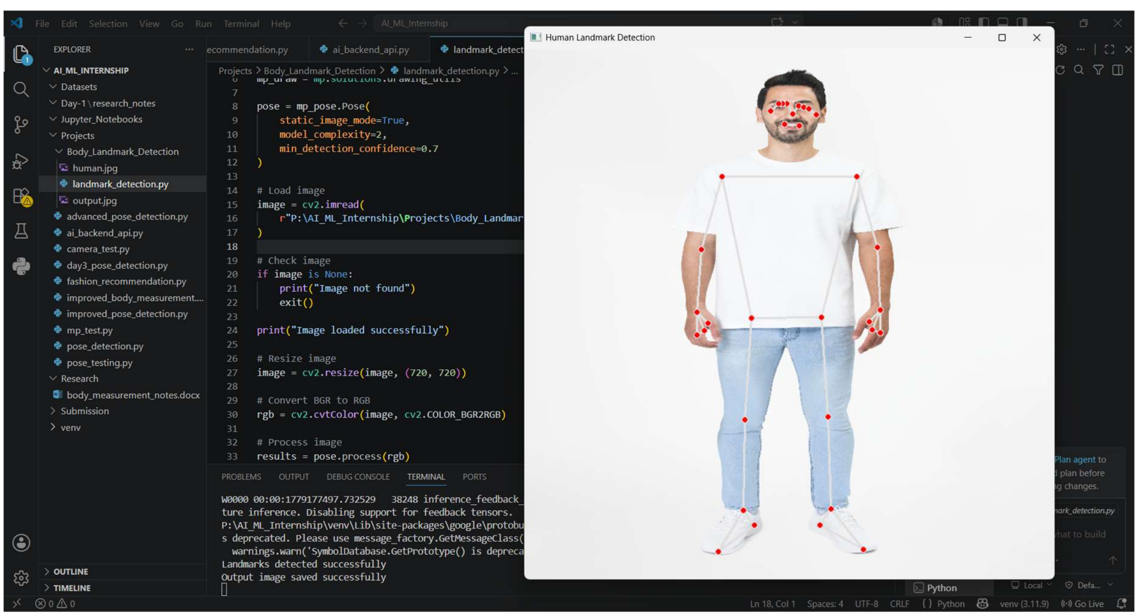



**Figure 3 — AI Landmark Testing Results**

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## 10. Challenges Faced

| Challenge | Impact |
|---|---|
| Poor lighting | Reduced landmark visibility |
| Fast body movement | Tracking instability |
| Complex backgrounds | Detection variation |
| Side body orientation | Minor pose estimation errors |

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## 11. Learning Outcomes

During this task, the following concepts were understood:

- human pose estimation
- AI landmark tracking
- skeletal visualization
- body posture analysis
- computer vision workflows
- MediaPipe pose detection
- AI body analysis systems

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## 12. Conclusion

The AI-Based Human Body Landmark Detection System successfully detected and tracked human body landmarks using Artificial Intelligence and Computer Vision technologies.

The task improved understanding of:

- AI pose estimation
- skeletal tracking systems
- landmark visualization
- body posture analysis
- AI body measurement logic
- computer vision applications

The system demonstrated how AI-based landmark detection technologies are used in:

- virtual fitting systems
- healthcare applications
- fitness tracking
- AI fashion technology
- posture analysis systems